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Graded Assessment - Cryptographic Tidbits

1. The Navajo Code Talkers were a relatively modern use of a cryptosystem in which

1 / 1 point

- ☐ a simple mathematical design produced a nearly unbreakable code.
- ☒ the spoken language was the key.
- ☐ The Caesar shift cipher.
- ☐ the Caesar shift cipher was used with different shift values.

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🟢 Correct
The Navajo language was unwritten and World War II had only a few dozen non-Navajo speakers in the world.

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2. The distinction between a 'code' and a 'cipher' is

1 / 1 point

- ☐ meaningless -- they are two terms for the same thing.
- ☐ that a cipher encrypts concepts while a code encrypts symbols.
- ☐ that a code is a binary file while a cipher is text message.
- ☒ that a code encrypts concepts while a cipher encrypts symbols.

🟢 Correct
Correct.

3. The term used for the art and science of breaking codes and ciphers is

1 / 1 point

- ☒ cryptanalysis.
- ☐ cryptology.
- ☐ social engineering.
- ☐ cryptography.

🟢 Correct
Correct.

4. In World War II, the primary military cipher used by the Germans in the field was

1 / 1 point

- ☐ The Vernam Cipher
- ☒ the Enigma machine
- ☐ The Vigenere Cipher
- ☐ JN-25

🟢 Correct
Correct.

5. The usual benchmark against which computational attacks are compared is

1 / 1 point

- ☒ the brute-force attack that systematically searches the key space.
- ☐ the social engineering attack.
- ☐ A chosen-plaintext attack.
- ☐ the best known attack against a similar cryptosystem.

🟢 Correct
Correct.

6. Which of the following, when used properly, is provably unbreakable?

1 / 1 point

- ☐ The Venona cipher
- ☒ The Vernam cipher
- ☐ Enigma
- ☐ The Vigenere cipher

🟢 Correct
This is also known as a One-Time Pad

7. Kerckhoff's Principle strongly recommends that

1 / 1 point

- ☒ it be assumed that the adversary has all information about the cryptosystem algorithms and its implementations.
- ☐ essential elements of the algorithm be kept private.
- ☐ implementation details remain closely guarded secrets.
- ☐ every detail about a cryptosystem be kept secret as much as possible.

🟢 Correct
The only thing the adversary lacks are the message keys.

8. The only piece of information that must be kept secret in most asymmetric cryptosystems is

1 / 1 point

- ☒ the private key.
- ☐ the underlying algorithm
- ☐ the public key.
- ☐ which asymmetric algorithm is being used in this exchange.

🟢 Correct
The private key is needed by only one party and need not ever be shared.

9. In most cryptosystems the distribution and verification of keys is generally accomplished by

1 / 1 point

- ☐ both parties performing some kind of 'key exchange' protocol.
- ☒ using a trusted third party
- ☐ using a bonded courier.
- ☐ publishing the public keys on the sender's webpage.

🟢 Correct
This is "Trent" in our Cast of Characters.

10. The name generally given to the actor than can insert messages into a communications channel is

1 / 1 point

- ☐ Alice
- ☐ Bob
- ☒ Malory
- ☐ Eve

🟢 Correct
Malory can alter or even insert entire messages.